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crystALL: Purely data-driven automated structure prediction for unidentified chemical compounds

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DOI: 10.xxxxxx/draft

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Submitted: 01 January 1970
Published: unpublished

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Statement of need

The ability to determine atomic structure of existing or hypothetical solids formed by chemical elements is one of the fundamental challanges in theoritical crystallography and in materials science as a whole. For the last several decades (Hawthorne, 1994), the two remaining challenges are predictions of (a) compositions forming compounds of interest and (b) candidate topologies or approximate atomic arrangements they can form. Given these, the exact structure can be solved experimentally, by matching X-ray diffraction profile to candidates using the method by (A. Le Bail et al., 1988) and interpreting the profile, or using deep learning (Yue et al., 2024). Similar analysis can be done, e.g., on the Raman spectrum (Ferrari & Basko, 2013). For hypothetical compounds in their pure states, the exact structure can also be solved using ab initio quantum mechanical methods, such as Kohn-Sham density functional theory (DFT) (Kohn & Sham, 1965), which are computationally expensive but can provide accurate results and continue to improve (Kothakonda et al., 2023).

The latter problem of proposing the topologies has been traditionally approached by finding geometries satisfying constraints, like the principles systematized by (Pauling, 1929), which included then-novel concepts like coördination. Over decades, this has evolved into automated computational tools like the `GRINSP` by (Armel Le Bail, 2005) which can propose such structures. More recently, increasing computational power shifted focus towards testing many options, with some of the methods starting from random arrangements and collapsing them into local minima, like `AIRSS` by (Pickard & Needs, 2011), or by evolutionary techniques, like `USPEX` (Glass et al., 2006).

Several efforts have been made to tackle both topological and compositional challenges simultaneously, such as high-throughput construction of the `AFLOW` (Curtarolo et al., 2012) DFT Database (Toher et al., 2018) (and others, see Methods Section) built by systematically populating the earlier, yet continuously growing, library of prototype structures (Mehl et al., 2016) with new chemistries based on the expert knowledge. However, even for a limited set of elements, this problem becomes combinatorically challenging (A. M. Krajewski et al., 2024). This prompted recent efforts into (1) systematic similarity-driven prediction of substitutions which would result in finding new low-energy structures (Wang et al., 2021); (2) into brute-force substitutions filtered by machine learning (ML) models (Schmidt et al., 2023); and (3) the combination thereof in (Ye et al., 2022) and (Merchant et al., 2023), leading to rapid discovery of new structures stable at 0K temperature and 0Pa pressure against competing arrangements.

While finding new candidates for zero-temperature zero-pressure hypothetical structures certainly opens the world for new discoveries, it warrants some key questions. Firstly, *"To what degree have we biased the discovery of new structures?"* through many iterations of biasing the search to our expectations, which is philosophically analogous to how retraining Large Language Models (LLMs) and other ML models on generated data makes them underestimate

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improbable events causing, so called, *Model Collapse* (Shumailov et al., 2023), and drift from the distribution of the real data (Alemohammad et al., 2023). In the context of materials, the latter is depicted well by Fig. 5 in (Ye et al., 2022) showing relatively small overlap between experimentally known structures and new ones predicted to be stable.

In case the data becomes biased towards low-energy solutions, one needs to ask *"Do we need the diverse energy landscapes to discover new materials?"* to which the answer is a multifaceted yes.

- Metastable are critical to fitting thermodyamic models as protodata. At the same time, CALPHAD does not care and can model unknown *phases* (Olson & Liu, 2023)
- Going beyond stability. It is not enough (Zunger, 2019) because of the complexities involved in the synthesis of new materials. Phase coexistance can take properties to extreme, e.g. infinitely negative thermal expansion (Liu et al., 2022). Superconductors (Ferreira et al., 2023)
- `AIRSS` generates polymorph populations like Fig. 6 in (Pickard & Needs, 2011) providing a diverse set of many metastable structures.

Summary

`crystALL` is a concise computational tool which leverages millions of previously calculated structures from authors' MPDD Database and other databases accessed through the OPTIMADE API (Evans et al., 2024) to identify structures of target stoichiometry and perform all permutations of substitutions to arrive at a list of candidate structures. This list

- (3 sentences) Past deployment on the Nd-Bi system (Im et al., 2022) and Figure 7a of (Im et al., 2022) showing excellent ordering of the candidates when evaluated vs different DFT methods. Past deployment in Al-Fe (Shang et al., 2021).
- Motivation slightly reiterating the summary with more focus on impact and need
- (**Rushi**, 3 sentences) *(Importance of Ru-Zn in the study of catalysts. Importance of KCl-LaCl3 system in the study of molten salts. How `crystALL` is continuously used in the lab to predict structures of unknown compounds and if the automated experience has been positive.)* To enhance nylon manufacturing efficiency, Ru-based catalysts are recognized as highly effective catalysts in synthesizing the key chemical intermediate through the benzene partial hydrogenation reaction (Zhang et al., 2013). Achieving precise control over this process is important to ensure optimal outcomes. Improvement of hydrogenation selectivity involves designing diverse active site ensembles in intermetallic catalysts (Dasgupta et al., 2022). Using crystALL, we have explored over 165, 000 configurations of Ru-Zn intermetallic compounds, including Ru_3Zn, RuZn, Ru_4Zn_9, $RuZn_3$, Ru_3Zn_{10}, $RuZn_6$, $RuZn_{10}$, and $RuZn_{13}$. We are currently examining their structures and stability through pySIPFENN (A. M. Krajewski et al., 2022), while simultaneously evaluating active site ensembles to design novel catalysts with high activity and selectivity. In the journey towards clean energy and carbon footprint reduction, the development of Molten Salt Reactors (MSRs) is essential, which requires high reliability and efficiency. crystALL serves as a valuable tool for identifying structures for complex molten salts, particularly in cases of limited or conflicting experimental data. For example, crystALL facilitated the prediction of structures for KLa_3Cl_{10} and $K_3La_5Cl_{18}$ compounds in the KCl-$LaCl_3$ system, thereby validating their stability reported in various literature sources (Seifert et al., 1985) (Song & Zheng, 1995).
- (**Adam**, 1 sentence) Conclusion on the importance of automation for user tool.

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Methods

MPDD Database

The MPDD or Materials-Properties-Descriptors Database is an infrastructure combining a growing collection of millions of atomic structures with a number of optimizations for rapid featurization and deployment of ML models. It includes author's own DFT data, as well as a number of other databases, including the OQMD (Shen et al., 2022), AFLOW (Toher et al., 2018), MP or Materials Project (Jain et al., 2013), NIST-JARVIS (Choudhary et al., 2020), COD (Gražulis et al., 2019), and GNoME (Merchant et al., 2023). The database is accessible through (a) a restricted high-throughput API serving enabling fetching of several thousand structures per second, and (b) a public OPTIMADE API (Evans et al., 2024) with low-throughput access but compatibility with other databases. Internally, the data is largely handled using modified pymatgen (Ong et al., 2013) kept up-to-date with upstream changes, to ensure wide compatibility with other tools in the field.

- optimade-python-tools (Evans et al., 2021)

Predictive Modeling Pipeline

(about the pySIPFENN, MPDD-ALIGNN, and other models)

- pySIPFENN software (A. M. Krajewski et al., 2022) and models on Zenodo (A. Krajewski et al., 2020)
- ALIGNN (Choudhary & DeCost, 2021)

(schematic of crystALL software)

Compound-Based Structure Prediction

(about the structure prediction)

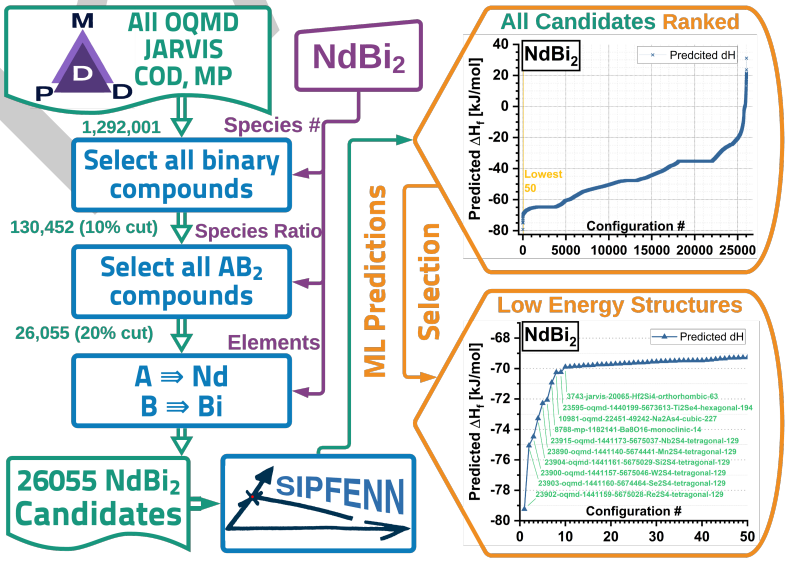


Figure 1: Compound-Based Structure Search. Reproduced from (Im et al., 2022) under permission.

Composition-Based Structure Prediction

(uncertainty in compound composition extends the previous method)

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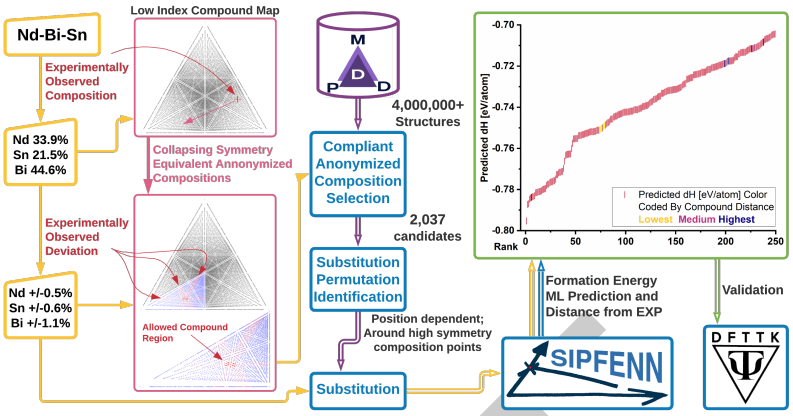


Figure 2: Composition-Based Structure Search.

Acknowledgements

- (a loooot of grants over 5 years: NSF superalloys, PSU Seed, all from Nd-Bi study, DOE ARPA-E, NSF-POSE, ONR FGM)
- All Phases Research Lab members who tested the software in production
- All Phases Research Lab members who provided feedback on the software

References

Alemohammad, S., Casco-Rodriguez, J., Luzi, L., Humayun, A. I., Babaei, H., LeJeune, D., Siahkoohi, A., & Baraniuk, R. G. (2023). *Self-consuming generative models go MAD*. https://doi.org/10.48550/arXiv.2307.01850

Bail, Armel Le. (2005). Inorganic structure prediction with GRINSP. *Journal of Applied Crystallography*, *38*, 389–393. https://doi.org/10.1107/S0021889805002384

Bail, A. Le, Duroy, H., & Fourquet, J. L. (1988). Ab-initio structure determination of LiSbWO6 by x-ray powder diffraction. *Materials Research Bulletin*, *23*, 447–452. https://doi.org/10.1016/0025-5408(88)90019-0

Choudhary, K., & DeCost, B. (2021). Atomistic line graph neural network for improved materials property predictions. *Npj Computational Materials*, *7*, 185. https://doi.org/10.1038/s41524-021-00650-1

Choudhary, K., Garrity, K. F., Reid, A. C. E., DeCost, B., Biacchi, A. J., Walker, A. R. H., Trautt, Z., Hattrick-Simpers, J., Kusne, A. G., Centrone, A., Davydov, A., Jiang, J., Pachter, R., Cheon, G., Reed, E., Agrawal, A., Qian, X., Sharma, V., Zhuang, H., … Tavazza, F. (2020). The joint automated repository for various integrated simulations (JARVIS) for data-driven materials design. *Npj Computational Materials*, *6*, 1–13. https://doi.org/10.1038/s41524-020-00440-1

Curtarolo, S., Setyawan, W., Hart, G. L. W., Jahnatek, M., Chepulskii, R. V., Taylor, R. H., Wang, S., Xue, J., Yang, K., Levy, O., Mehl, M. J., Stokes, H. T., Demchenko, D. O., & Morgan, D. (2012). AFLOW: An automatic framework for high-throughput materials discovery. *Computational Materials Science*, *58*, 218–226. https://doi.org/10.1016/j.commatsci.2012.02.005

Dasgupta, A., He, H., Gong, R., Shang, S.-L., Zimmerer, E. K., Meyer, R. J., Liu, Z.-K., Janik, M. J., & Rioux, R. M. (2022). Atomic control of active-site ensembles in ordered alloys to enhance hydrogenation selectivity. *Nature Chemistry*, 1–7. https://doi.org/10.1038/

DRAFT

s41557-021-00855-3

Evans, M. L., Andersen, C. W., Dwaraknath, S., Scheidgen, M., Fekete, Á., & Winston, D. (2021). 'Optimade-python-tools': A python library for serving and consuming materials data via OPTIMADE APIs. *Journal of Open Source Software*, *6*, 3458. https://doi.org/10.21105/JOSS.03458

Evans, M. L., Bergsma, J., Merkys, A., Andersen, C. W., Andersson, O. B., Beltrán, D., Blokhin, E., Boland, T. M., Balderas, R. C., Choudhary, K., Díaz, A. D., García, R. D., Eckert, H., Eimre, K., Montero, M. E. F., Krajewski, A. M., Mortensen, J. J., Duarte, J. M. N., Pietryga, J., … Armiento, R. (2024). *Developments and applications of the OPTIMADE API for materials discovery, design, and data exchange*. https://doi.org/10.48550/arXiv.2402.00572

Ferrari, A. C., & Basko, D. M. (2013). Raman spectroscopy as a versatile tool for studying the properties of graphene. *Nature Nanotechnology*, *8*, 235–246. https://doi.org/10.1038/nnano.2013.46

Ferreira, P. P., Conway, L. J., Cucciari, A., Cataldo, S. D., Giannessi, F., Kogler, E., Eleno, L. T. F., Pickard, C. J., Heil, C., & Boeri, L. (2023). Search for ambient superconductivity in the lu-n-h system. *Nature Communications*, *14*, 5367. https://doi.org/10.1038/s41467-023-41005-2

Glass, C. W., Oganov, A. R., & Hansen, N. (2006). USPEX—evolutionary crystal structure prediction. *Computer Physics Communications*, *175*, 713–720. https://doi.org/10.1016/j.cpc.2006.07.020

Gražulis, S., Merkys, A., Vaitkus, A., Chateigner, D., Lutterotti, L., Moeck, P., Quiros, M., Downs, R. T., Kaminsky, W., & Bail, A. L. (2019). Crystallography open database: History, development, and perspectives. In *Materials Informatics* (pp. 1–39). Wiley. https://doi.org/10.1002/9783527802265.ch1

Hawthorne, F. C. (1994). Structural aspects of oxide and oxysalt crystals. *Acta Crystallographica Section B Structural Science*, *50*, 481–510. https://doi.org/10.1107/S0108768193014466

Im, S., Shang, S. L., Smith, N. D., Krajewski, A. M., Lichtenstein, T., Sun, H., Bocklund, B. J., Liu, Z. K., & Kim, H. (2022). Thermodynamic properties of the nd-bi system via emf measurements, DFT calculations, machine learning, and CALPHAD modeling. *Acta Materialia*, *223*, 117448. https://doi.org/10.1016/J.ACTAMAT.2021.117448

Jain, A., Ong, S. P., Hautier, G., Chen, W., Richards, W. D., Dacek, S., Cholia, S., Gunter, D., Skinner, D., Ceder, G., & Persson, K. A. (2013). Commentary: The materials project: A materials genome approach to accelerating materials innovation. In *APL Materials* (Vol. 1, p. 011002). American Institute of Physics Inc. https://doi.org/10.1063/1.4812323

Kohn, W., & Sham, L. J. (1965). Self-consistent equations including exchange and correlation effects. *Physical Review*, *140*, A1133–A1138. https://doi.org/10.1103/PhysRev.140.A1133

Kothakonda, M., Kaplan, A. D., Isaacs, E. B., Bartel, C. J., Furness, J. W., Ning, J., Wolverton, C., Perdew, J. P., & Sun, J. (2023). Testing the r2SCAN density functional for the thermodynamic stability of solids with and without a van der waals correction. *ACS Materials Au*, *3*, 102–111. https://doi.org/10.1021/ACSMATERIALSAU.2C00059/SUPPL_FILE/MG2C00059_SI_002.PDF

Krajewski, A. M., Beese, A. M., Reinhart, W. F., & Liu, Z.-K. (2024). *Efficient generation of grids and traversal graphs in compositional spaces towards exploration and path planning exemplified in materials*. https://doi.org/10.48550/arXiv.2402.03528

Krajewski, A. M., Siegel, J. W., Xu, J., & Liu, Z. K. (2022). Extensible structure-informed prediction of formation energy with improved accuracy and usability employing neural networks. *Computational Materials Science*, *208*. https://doi.org/10.1016/J.COMMATSCI.

DRAFT

2022.111254

Krajewski, A., Siegel, J., Xu, J., & Liu, Z.-K. (2020). *Neural networks for structure-informed prediction of formation energy (employed in SIPFENN)*. Zenodo. https://doi.org/10.5281/zenodo.4006802

Liu, Z.-K., Wang, Y., & Shang, S.-L. (2022). Zentropy theory for positive and negative thermal expansion. *Journal of Phase Equilibria and Diffusion*. https://doi.org/10.1007/s11669-022-00942-z

Mehl, M. J., Hicks, D., Toher, C., Levy, O., Hanson, R. M., Hart, G., & Curtarolo, S. (2016). *The AFLOW library of crystallographic prototypes*. http://arxiv.org/abs/1607.02532

Merchant, A., Batzner, S., Schoenholz, S. S., Aykol, M., Cheon, G., & Cubuk, E. D. (2023). Scaling deep learning for materials discovery. *80 | Nature |, 624*. https://doi.org/10.1038/s41586-023-06735-9

Olson, G. B., & Liu, Z. K. (2023). Genomic materials design: CALculation of PHAse dynamics. *Calphad*, *82*, 102590. https://doi.org/10.1016/j.calphad.2023.102590

Ong, S. P., Richards, W. D., Jain, A., Hautier, G., Kocher, M., Cholia, S., Gunter, D., Chevrier, V. L., Persson, K. A., & Ceder, G. (2013). Python materials genomics (pymatgen): A robust, open-source python library for materials analysis. *Computational Materials Science*, *68*, 314–319. https://doi.org/10.1016/j.commatsci.2012.10.028

Pauling, L. (1929). The principles determining the structure of complex ionic crystals. *Journal of the American Chemical Society*, *51*, 1010–1026. https://doi.org/10.1021/ja01379a006

Pickard, C. J., & Needs, R. J. (2011). Ab initio random structure searching. *Journal of Physics: Condensed Matter*, *23*, 053201. https://doi.org/10.1088/0953-8984/23/5/053201

Schmidt, J., Hoffmann, N., Wang, H.-C., Borlido, P., Carriço, P. J. M. A., Cerqueira, T. F. T., Botti, S., Marques, M. A. L., Schmidt, J., Hoffmann, N., Wang, H.-C., Marques, M. A. L., Borlido, P., Carriço, P. J. M. A., Cerqueira, T. F. T., & Botti, S. (2023). Machine-learning-assisted determination of the global zero-temperature phase diagram of materials. *Advanced Materials*, *35*, 2210788. https://doi.org/10.1002/ADMA.202210788

Seifert, H. J., Fink, H., & Thiel, G. (1985). Thermodynamic properties of double chlorides in the systems ACl/LaCl3 (A Na, K, Rb, Cs). *Journal of The Less-Common Metals*, *110*(1-2), 139–147. https://doi.org/10.1016/0022-5088(85)90315-7

Shang, S.-L., Sun, H., Pan, B., Wang, Y., Krajewski, A. M., Banu, M., Li, J., & Liu, Z.-K. (2021). Forming mechanism of equilibrium and non-equilibrium metallurgical phases in dissimilar aluminum/steel (al–fe) joints. *Scientific Reports*, *11*, 24251. https://doi.org/10.1038/s41598-021-03578-0

Shen, J., Griesemer, S. D., Gopakumar, A., Baldassarri, B., Saal, J. E., Aykol, M., Hegde, V. I., & Wolverton, C. (2022). Reflections on one million compounds in the open quantum materials database (OQMD). *Journal of Physics: Materials*, *5*, 031001. https://doi.org/10.1088/2515-7639/ac7ba9

Shumailov, I., Shumaylov, Z., Zhao, Y., Gal, Y., Papernot, N., & Anderson, R. (2023). *The curse of recursion: Training on generated data makes models forget*. https://doi.org/10.48550/arXiv.2305.17493

Song, X., & Zheng, C. (1995). An Investigation on the Phase Diagram of Ternary System LaCl3-KCl-LiCl in Chinese). *Acta Chimica Sinica*, *53*, 978–984.

Toher, C., Oses, C., Hicks, D., Gossett, E., Rose, F., Nath, P., Usanmaz, D., Ford, D. C., Perim, E., Calderon, C. E., Plata, J. J., Lederer, Y., Jahnátek, M., Setyawan, W., Wang, S., Xue, J., Rasch, K., Chepulskii, R. V., Taylor, R. H., … Curtarolo, S. (2018). The AFLOW fleet for materials discovery. In *Handbook of Materials Modeling* (pp. 1–28). Springer

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International Publishing. https://doi.org/10.1007/978-3-319-42913-7_63-1

Wang, H.-C., Botti, S., & Marques, M. A. L. (2021). Predicting stable crystalline compounds using chemical similarity. *Npj Computational Materials*, *7*, 12. https://doi.org/10.1038/s41524-020-00481-6

Ye, W., Lei, X., Aykol, M., & Montoya, J. H. (2022). Novel inorganic crystal structures predicted using autonomous simulation agents. *Scientific Data*, *9*, 302. https://doi.org/10.1038/s41597-022-01438-8

Yue, W., Tripathi, P. K., Ponon, G., Ualikhankyzy, Z., Brown, D. W., Clausen, B., Strantza, M., Pagan, D. C., Willard, M. A., Ernst, F., Ayday, E., Chaudhary, V., & French, R. H. (2024). Phase identification in synchrotron x-ray diffraction patterns of ti–6Al–4V using computer vision and deep learning. *Integrating Materials and Manufacturing Innovation*, *13*, 36–52. https://doi.org/10.1007/s40192-023-00328-0

Zhang, P., Wu, T., Jiang, T., Wang, W., Liu, H., Fan, H., Zhang, Z., & Han, B. (2013). Ru–Zn supported on hydroxyapatite as an effective catalyst for partial hydrogenation of benzene. *Green Chemistry*, *15*(1), 152–159. https://doi.org/10.1039/C2GC36596K

Zunger, A. (2019). Beware of plausible predictions of fantasy materials. *Nature 2021 566:7745*, *566*, 447–449. https://doi.org/10.1038/d41586-019-00676-y

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